

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/367090586>

The Evolution of Cloud Computing

Preprint · January 2023

CITATIONS

0

READS

3,841

4 authors, including:



Babita Bhagat

Pillai Hoc College Of Engineering And Technology

8 PUBLICATIONS 24 CITATIONS

SEE PROFILE



Shrutika Tularam Khobragade

Pillai Hoc College Of Engineering And Technology

6 PUBLICATIONS 2 CITATIONS

SEE PROFILE

The Evolution of Cloud Computing

Babita Bhagat*, Shrutika Khobragade, Archana Arudkar

Abstract

This study comprehensively covers cloud computing; the diverse help models, its development, how could it be significant against the background of the arising Indian market and how might this innovation be tapped with the goal that its latent capacity is utilized to limit with regards to mechanical development in the country. Distributed computing as an idea has been around for a long while; however, the new years have seen it being used in different backgrounds considering the different benefits it has. India is bit by bit turning into the trail blazer being used and proliferation of cloud innovation. This research work targets laying out the essence of this innovation, the key qualities and to search for territories of progress where the innovation can be utilized and can be improved later on.

Keywords: Cloud computing, history, PaaS, IaaS, SaaS, evolution, future in India

INTRODUCTION

Accepting distributed computing arrangements is changing into continuously prevailing for associations attempting to drive their business forward throughout the moving toward years, with the parts of CTOs and CIOs developing in sync. albeit "computerized change" keeps on being the monstrous hogwash of the moment, with inclusion of (AI), AI (ML), and virtual and expanded reality (VR and AR) expanding as for quite a while ticks by Buyya *et al.* [1], cloud change stays current but it has slipped from the total glare of the spotlight inside the norm on-line press. In the background, nonetheless, distributed computing keeps on being a hotly debated issue. Indeed, it is presently not a matter of whether or not a company can bear a cloud change [2]. Or maybe, inquiries on precisely what administrations and tasks should be moved into the cloud, what amount would they be able to cost, what amount would they be able to save the business, and the way rapidly will everything occur, square measure ruling the communicated in language. However, any place has the distributed computing unrest got back from? What square measure causes, any place square measure we keep an eye on these days, and where's it going straightaway? We should consider the set of experiences and development of distributed computing that has brought the United States up to the current, and analyze the patterns which will affect the more drawn out term.

A Brief History of Cloud Computing

Right when we consider the cloud, inconsistently do we cast our minds back to times before the 21st century. In light of everything, it has genuinely been an absurd decade or so that circulated figuring really started to frame into the goliath, universal and all-astounding behemoth we know today as shown in Figure 1 [3].

*Author for Correspondence

Babita Bhagat
E-mail: bbhagat@mes.ac.in

Assistant Professor, Department of Computer Engineering,
Pillai HOC College of Engineering & Technology, Rasayani,
Maharashtra, India

Received Date: April 15, 2021
Accepted Date: April 20, 2021
Published Date: April 24, 2021

Citation: Babita Bhagat, Shrutika Khobragade, Archana Arudkar. The Evolution of Cloud Computing. Journal of Network Security. 2021; 9(1): 20–28p.

In any case, real thoughts of the cloud have existed for a few, various years, and believe it or not can be followed as far back as the 1950s with incorporated workers preparing. In those early days, concentrated worker PCs were titanic machines, and incredibly, exorbitant, too expensive to even think about in the evening considering buying and saving one for every single agent.

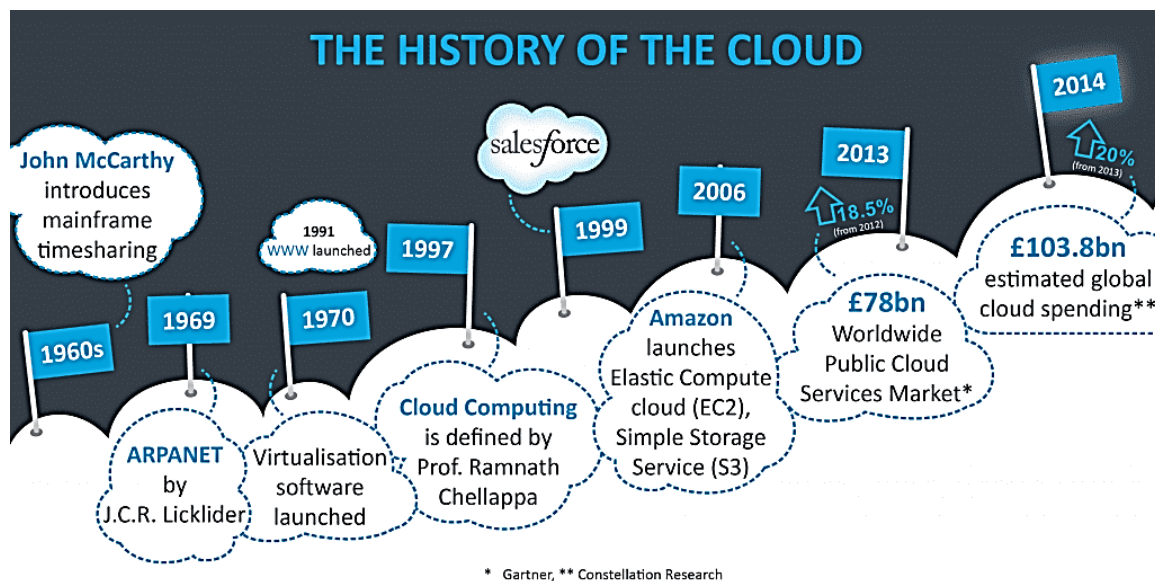


Figure 1. Cloud computing evolution phases [4].

Also, clearly, not many out of each odd single specialist expected induction to one reliably as they do today. Taking everything into account, most affiliations would be two or three machines, and thereafter realize "time-sharing" plans which engaged various customers to get to the central concentrated worker PC from related stations. These stations were known as "inept terminals", and gave no dealing with power of their own. In light of everything, this sort of shared computational power is the fundamental, essential explanation of appropriated registering, and where everything began [5].

During the 1960s, a critical movement in circulated figuring came when American PC analyst J.C.R. Licklider conceptualized an interconnected course of action of PCs. In 1969, "Lick", as he is much of the time known, developed an unrefined interpretation of the Internet, known as the Advanced Research Projects Agency Network (ARPANET). ARPANET was the primary association that allowed progressed sources to be split between PCs that were not in a comparable genuine territory. Lick's vision was moreover for a presence where everyone would be interconnected through PCs and prepared to get to information from wherever. Sound unmistakable? Clearly it does; it is the Internet, all things considered, and a requirement for getting to every one of the benefits that the cloud sorts it out.

All through the drawn out that followed, various further movements in cloud advancement showed up. Consider an example, in 1972, IBM conveyed a working structure (OS) called the Virtual Machine (VM) working system. Virtualization is the one which is a virtual PC that acts similarly as a certifiable one, and acts with an operational OS. The thought progressed with the Internet, and associations began offering "virtual" private associations which can be rented as assistance, in the end inciting the improvement of the bleeding edge conveyed processing establishment during the 1990s.

Moreover, in this decade, media correspondences associations began offering virtualized private associations, which had a comparable assistance quality as their submitted feature point data relationship at a reduced cost. Instead of working out real structure to consider more customers to have their own affiliations, media interchanges associations were right now prepared to give customers shared induction to a comparative real establishment [6].

During the year of 2000s, Amazon Web Services (AWS) emerged, and Amazon dispatched Elastic Compute Cloud (EC2) in 2006, allowing associations and individuals to buy space and rent virtual

PCs through which everyone can make use of their own undertakings as well as applications. In the specific year, Google dispatched its Google Docs organizations, allowing customers to save, adjust and move reports in the cloud.

In 2007, IBM, Google, and a couple of universities joined to develop a laborer farm for research projects. It was furthermore the year where applications like Netflix dispatched its video electronic element, by the cloud to send movies and other video content into the homes and onto the PCs of all around the world (and in the end countless) allies around the globe.

DISTRIBUTED COMPUTING

There are various help contributions and usage models under the haze figuring umbrella. The NIST meaning of distributed computing characterizes three conveyance models.

- Software-as-a-Service (SaaS);
- Platform-as-a-Service (PaaS); and
- Infrastructure-as-a-Service (IaaS).

Various Layers in Cloud Computing

In conveyance models of cloud computing, the cloud management providers can suggest organizations at different cloud layers of the resource stack, recreating the limits done by the applications, working systems, or genuine hardware [7].

- *Software as a Service (SaaS)*: It provides finished applications that end users as well as customers can access through a feeble client (routinely, anyway not actually, a web program). Observable cases of SaaS consolidate various applications like Gmail, Google drive, Google Docs, and Salesforce.com. The end users as well as customers do not rehearse any order over the arrangement of the application (close to some minute changes as well as customization and plan decisions), laborers, frameworks organization, and limit establishment.
- *Platform as a Service (PaaS)*: It offers a working structure similarly as set-ups of programming vernaculars and programming headway instruments that customers will develop and use applications very easily. Perceptible models consolidate Microsoft Windows Azure and Google App Engine. PaaS gives end users and customers command over many application arrangements yet does not give them authority done by the genuine establishment.
- *Infrastructure as a Service (IaaS)*: It provides end customers direct permission to planning, storing and other handling resources and grants them to orchestrate those various resources as well as run working systems and programming on them as they see fit. Occurrences of IaaS fuse Amazon Elastic Compute Cloud (EC2), Rackspace, and IBM Computing on Demand. Service Deployment Models of The National Institute of rules and development (NIST) definition portrays four sending models: Private cloud, Community cloud, Public cloud, and Hybrid cloud.
- *Private Cloud*: This cloud establishment just used by a singular relationship with different claims to fame units. It very well may be administered, worked, and guaranteed by the affiliation or an untouchable or a blend of them.
- *Community Cloud*: This cloud structure is shared by a couple of affiliations. It supports a specific neighbourhood and has shared concerns (e.g., mission, security necessities, system, and consistent thoughts). It may be guaranteed, managed, and worked by at any rate, one of the affiliations locally, an untouchable, or a blend of them.
- *Public Cloud*: This cloud system is open to the general populace or a tremendous industry get-together and it very well may be asserted, supervised, and worked by a business, educational, or government affiliation, or a blend of them.
- *Hybrid Cloud*: It consists of a mixture, at any rate two fogs (public, private or neighbourhood). It stays a novel substance yet is bound together. Services offered by cloud organizations that can be passed on using circulated figuring, through good situations of the scattered cloud model.

-
- *Hosted Desktops*: An encouraged work region acts and takes after a standard work region PC, anyway the item and data customers using are housed in far away, significantly secure worker ranches, rather than running on its owned machines.
 - *Hosted Email*: However emails are taken care of mostly on managed labourers, giving redundancy and brisk organization from any zone. This licenses customers to get to their email, timetable, contacts and shared records by a grouping of means.
 - *Voice over IP (Hosted Telephony)*: It is a technique for passing on calls and organizations across modernized web associations.
 - *Cloud Storage*: It is the movement of data amassing as a help, from a pariah provider, with access through the web and charging decided on limit used in a particular period.
 - *Dynamic Servers*: However, a provider like Think Grid gives all customer induction to resources that look and feel definitely like a given labourer, anyway that are totally flexible. As per the present side of market status, cloud enrolling continues getting standard determination as more associations migrate in the cloud. Market advancement in different kinds of cloud organization models is given under. SaaS Market: There customer relationship management, Financial Planning, HR, Word getting ready kind of uses are provided by the providers. Email benefits furthermore come in SaaS arrangement. Some SaaS expert communities are given underneath.
 - *Salesforce.com*: It gives ideas of call centers, the board, complaint following, and organization entrance organizations. • Email cloud: Email cloud is a remarkable email illuminating expert association. These organizations save customer time, information transmission and the issue of overseeing email advising issues like spam, diseases, individual time and moving organizations.
 - *Google Apps*: It is an assistance from Google giving uninhibitedly versatile variations of a couple of Google things under a custom space name. It incorporates a couple of Web applications with the same kind of value to standard office suites, including: Gmail, Google Groups, Google Calendar, Google DocsTalk, and Sites. Up to walk, 2010 around 25 million people had "changed to Google Apps" According to Google sites. PaaS market: There application plan, application progression, testing, sending, and encouraging similarly as application organizations, for instance, bunch joint exertion, web organization coordination and marshalling, database compromise, security, adaptability, accumulating, steadiness, express the heads, application framing, application instrumentation and fashioner neighbourhood sort of organizations gave. Some PaaS expert centers are given underneath.
 - *Google AppEngine*: It is centered around just standard web applications, approving an application plan of clean segment between a stateless estimation level and a stateful amassing level. AppEngine's extraordinary modified scaling and high openness frameworks and the select MegaStore of data storing open to AppEngine applications, all rely upon these impediments.
 - *Microsoft's Azure*: It is made using the .NET libraries, and organized to the Common Language Runtime, a language-independent directed environment. Thus, Azure is momentary between various application frameworks like AppEngine and gear virtual machines like EC2 is the provider of appropriate processing as PaaS.
 - *Rackspace*: The Rackspace Cloud was initially dispatched as Mosso LLC, an entirely claimed auxiliary start-up charged for the utility processing offering. It is also known as a web application facilitating/cloud stage supplier ("Cloud Sites") that bills on a utility processing premise. It has been stretched out into distributed storage ("Cloud Files"). It was one of the main business distributed computing administrations.
 - *AWS: S3: vCloud* is a distributed computing activity from VMware and it will permit clients to relocate work on request from their "inside cloud" of collaborating VMware hypervisors to a far off haze of VMware hypervisors. The objective of the activity is to furnish the force of distributed computing with the adaptability permitted by virtualization. IaaS market: The assistance offers numerous choices for figuring, memory, network setup, working framework
-

and Disaster Recovery (DR) to accommodate customer's particular necessities. A few IaaS specialist co-ops are given beneath.

- *Flexiscale*: FlexiScale is a finished reconstruction of Europe's first distributed computing stage utilizing Flexiant's progressive Extility innovation.
- *AWS*: Amazon's EC2 conveys versatile, pay-more only as costs arise register limit in the cloud. Amazon EC2 is toward one side of the range. An EC2 occurrence looks similar as actual equipment, and clients can handle almost the whole programming stack, from the part down to up.
- *NetMagic Solutions*: These supplier groups screen all basic boundaries identified with the presentation of the organization and the workers being facilitated.
- *InstaCompute which is from Tata Communications*: It gives pre-or-post-paid instalment alternatives. Different preparing power choices. It can be private or may be public connectivity. InstaCompute changes with our evolving prerequisites.

Virtualization

- Web 2.0.
- World-wide appropriated stockpiling framework • Distributed Computing.
- Grid Computing [8].
- Utility Computing.
- Network Bandwidth and Latency.
- Fault-Tolerant Systems.
- Programming Models.

SECURITY AND PRIVACY

Cloud computing postures security concerns since the benefit supplier can get to the information that is within the cloud at any time. It seems to inadvertently or intentionally change or erase information. Various cloud suppliers can share data with third parties in case vital for purposes of law and arrange without a warrant. That is what is allowed in the security approaches, which clients must concur to; sometime recently, they begin utilizing cloud administrations. Arrangements to security incorporate arrangement and enactment as well as end-users' choices for how all this information is stored. Clients can scramble information that is prepared or put away inside the cloud to anticipate unauthorized access. Personality administration frameworks can too give viable arrangements to protection concerns in cloud computing. These systems are a way to distinguish between authorized and unauthorized clients and decide the sum of information that is open to each entity. The frameworks work by making and depicting personalities, recording exercises, and getting freed of identities which are not used.

Agreeing to the cloud security collusion, the best three dangers within the cloud are: Uncertain Interfacing and APIs, Information Misfortune and Spillage, and Equipment Failure; which accounted for 29, 25 and 10% of all cloud security blackouts separately. Together, these shape shared development weaknesses. In a cloud provider stage being shared by particular customers, there might be a believability that information having a spot to distinctive client's dwells on the same information server. Also, Eugene Schultz, chief innovation officer at Emagined Security, said that programmers are investing substantial time and exertion seeking out for ways to enter the cloud. "There are a few genuine Achilles' heels within the cloud foundation that are making enormous gaps for the awful folks to urge in to". Since information from hundreds or thousands of companies can be put away on huge cloud servers, programmers can hypothetically pick up control of tremendous stores of data through a single attack, a prepare he called "hyperjacking". A few cases of this incorporate the Dropbox security breach, and iCloud 2014 leak. Dropbox had been breached in October 2014, having over 7 million of its clients' passwords stolen by programmers in an exertion to urge money related esteem from it by Bitcoins (BTC). By having these passwords, they are able to examine private information as well as have this information be recorded by look motors (making the data public).

There is the issue of legitimate possession of the information (On the off chance that a client stores some information within the cloud, can the cloud supplier benefit from it?). Numerous terms of benefit understandings are noiseless on the address of ownership. Physical control of the computer gear (private cloud) is more secure than having the hardware off-site and beneath somebody else's control (open cloud). This conveys incredible motivating force to open cloud computing benefit suppliers to prioritize building and keeping up solid administration of secure services. A few little businesses that do not have ability in IT security seem to discover that it is more secure for them to utilize an open cloud.

There is the hazard that end clients do not get the issues included when marking on to a cloud benefit (people some of the times do not study the numerous pages of the terms of benefit understanding, and just click "Acknowledge" without perusing). This is often imperative presently that cloud computing is getting to be well known and required for a few administrations to work, for illustration for a shrewdly individual right hand (Apple's Siri or Google Presently). In a general sense, private cloud is seen as more secure with higher levels of control for the proprietor, be that as it may open cloud is seen to be more adaptable and requires less time and cash venture from the user.

LIMITATION AND DISADVANTAGES

Agreeing to Bruce Schneier, "The drawback is simply will have constrained customization choices. Cloud computing is cheaper since of financial matters of scale, and like any outsourced task, you tend to urge what you need. An eatery with a constrained menu is cheaper than an individual chef who can cook anything you need. Less choices at a much cheaper cost: it's an include, not a but". The paper author also suggests that "the cloud supplier might not meet your lawful needs" which businesses got to weigh the benefits of cloud computing which is against the risks. In cloud computing, the control of the cloud which is the back conclusion foundation is constrained to the cloud merchant as it were. The suppliers frequently choose on the administration approaches, which is moderated by what the cloud clients are able to do with their deployment. Cloud clients are too restricted to the control and administration of their applications, all the information and services [9].

This incorporates all the information caps, which are there on the cloud clients. So, by the cloud seller apportioning a certain sum of transmission capacity for each client and are regularly shared among other cloud users. Privacy and secrecy are huge concerns in a few exercises. For occasion, sworn interpreters working beneath the stipulations of an NDA might confront issues with respect to touchy information which is not in an encrypted form. Cloud computing is advantageous to numerous ventures; it brings down costs and permits them to center on competence rather than on things of IT and framework. By the by, cloud computing has demonstrated to have a few restrictions and impediments, particularly for littler commerce operations, especially with respect to security and downtime. Specialized blackouts are unavoidable and happen now and then when cloud benefit providers (CSPs) end up overpowered within the preparation of serving their clients.

This may result in transitory trade suspension. Since this technology's frameworks depend on the Web, a person cannot get to their applications, server, or information from the cloud amid an outage. Be that as it may, numerous huge endeavors keep up at slightest two web suppliers, utilizing distinctive section focuses into their work environments, a few indeed utilize 4G as a third fallback.

FUTURE OF CLOUD COMPUTING IN INDIA

One of the striking features of appropriated registering is that it presents benefits to each piece of Indian economy including anyway not limited to SMEs, the preparation system, and the Government of India. Circulated processing is transforming into all unavoidable and when executed identified with versatile development, it sets stage for having power of cloud with the convenience of transportability. This model is currently coming into picture with applications. Possible destiny of appropriated registering lies in making the cloud model goes absolutely on pay all the more just as expenses emerge premise for large business applications just as for applications in handheld

contraptions. The assessment of the cloud organizations market in India is needed to hit a record high of more than \$556 million preceding the completion of 2014. Advancement for the year is projected at 32.2%, which will see India's cloud region climb from \$421 million out of 2013 to \$556.8 this year, an extension of \$135.8 million. This high speed of addition is dependent upon to hold for the accompanying very few years, and by 2018, the Indian cloud organizations market could hit a vacillating \$1.7 billion on the whole out worth. The numbers moreover show that Infrastructure-as-a-Service (IaaS) will experience a great deal of improvement in India, creating from \$58 million a year prior to \$78 million this year (a differentiation of \$20 million, or 34.5%). As cloud organizations grow, so will IaaS, as shown by projections, which will hit around \$317 million by 2018. Related portions like Business Process-as-a-Service (BPaaS) and Software-as-a-Service (SaaS) will develop, too. BPaaS will create from \$62.3 million to \$75 million this year (\$12.7 million, or 20.4%), and SaaS, which will experience the most advancement by and large by 2018, will hit \$636 million preceding that year's finish. BPaaS will create \$204 million preceding the completion of 2018, setting it in third spot behind SaaS and IaaS. India is accepting cloud-based systems to a steadily expanding degree, as they require less genuine additional room, are more cost gainful, and strong. There are stresses over assurance and receptiveness; anyway, determination of the advancement keeps on extra high. Other assessment shows private cloud scene in India evaluated that the total cloud market in India, at present at \$400 million, will show up at a market assessment of \$ 4.5 billion by 2015. Of this, private cloud choice will overpower and address \$3.5 billion in wages, creating at over 60%. The examination moreover evaluates that the segment will make 1 lakh occupations by 2015 from 10,000 today. With the overall environment of cloud choice snappy creating in India, conveyed figuring will address a basic proposal in the outright IT spends of endeavors. Complete fogs spend as a level of the hard and fast IT spends is dependent upon to climb from 1.4% in 2010 to 8.2% in 2015. IT/ITeS, telecom, BFSI, gathering and government territories will contribute the greatest to the cloud market in India, with practically 78% of the hard and fast market. Along with generally government bodies, the Indian government has moreover taken an unmistakable interest for the advancement and is needed to create at a quick speed of 45.1% yearly until 2018 contributing an amount of \$18 billion. India has seen different forefront firms from the US secure its more unobtrusive IT associations because of their work with cloud-based headways. New organizations in Bangalore, Hyderabad, and Pune have been gotten by moderate size associations and tremendous organizations the equivalent throughout the latest year as IT firms continue pushing towards growing their circulated registering limits. The creating number of Internet customers and extension in phone gathering in India is an imperative factor for its cloud advancement, neither of which gives a great deal of sign of moving down throughout the span of the accompanying very few years [10]. Apart from the numbers, the cloud development is endlessly progressing in the features and designing models giving more keen game plans and structure flexibility to new organizations, SMEs and IT associations yet also government bodies and workplaces. The Indian government is moreover developing a cloud procedure, to be explicit the GI cloud (or Meghraj) which will see more prominent clarity in the near future. Wide augmentation of moderate broadband accessibility (and mobile phones nearby) is needed to give further open entryways in e-Governance and transportability courses of action. The Indian cloud is on the rising, yet there are certain movements that are ordinary and Cloud is creating more than ever. What about we research the possible future unforeseen developments.

Proactive Application Monitoring

Proactive application noticing development is as of now available, anyway farsighted advancement and programming will in a little while make this more solid and definite. Associations will really need to anticipate catastrophe and redirect it, directing mischief to their systems. This will hinder excursion and make the association safer.

Innovation to Ensure Uptime

With low-power processors, worker homesteads will end up being more moderate, allowing associations to get seven to ten worker ranches around the world in different time locales and thus allowing them to guarantee 99.9% uptime. This will keep associations away from losing money and

falling prey to their opponents. Windows Azure as of now is giving 99.99% SLA by giving data abundance across various worker ranches in the world

Capacity to Validate Identities through Trusts

Later on, cloud security systems will really need to favor characters through a "united trust". Identity-based security is accepted to be more secure than current kinds of safety. More people will begin to accept dispersed figuring when this happens.

Versatile Optimization for Cloud Services

Versatile exchange is on the rising. All the registering applications which require less resources and are proposed for phones. Accessibility is extended considering the way that less number of resources are required. This is the explanation of dispersed processing and is proposed for business and all independent applications.

Distributed Computing will Help Associations with Smoothing out their Hypotheses

Cloud figuring can help associations with redesigning adventures and scale exercises. Later on, remarkable disappointment of all the power processors and some of the other chip development will help associations with working their full IT limit. With greater turn of events, more imperative pay potential is possible. Associations can re-put assets into their things as well as in organizations with higher generally speaking incomes. Scaling theories will similarly provoke more conspicuous operational deftness.

Information Recovery and Remote Access

Cloud preparing engages and improves removed admittance and snappier disaster recovery. Right when associations have an emergency information security system with security entrance tests, associations can keep up their genuine edge inside their different endeavors. With conveyed processing, a couple of associations that did not see an infiltrate may recover inside the space of minutes as opposed to hours. Losing prohibitive data can impair an association and even explanation approaches to close. Every association should move to dispersed registering consequently.

Unified Data

Unified data is ordinary in abundance later on for dispersed figuring. This licenses associations to make huge informational indexes. Persevering thought can be improved with united data in gigantic informational collections. Better monetary trade decisions are similarly possible. Concentrated data is a strategy for what might be on the horizon.

Upgraded Connectivity

Portable devices that rely upon the cloud will end up being every one of the more amazing and more slim because all these applications will be on the web. All phones will keep data that stays in the cloud, and all the designers can be added to more prominent limits and it also reduces the costs of the phone. The one outline of this thought is Apple's iCloud.

Upgraded Security

Information center genuine security is likewise pretty much as huge as data encryption. While least encryption rules may be extended from the current 256-digit SSL, real induction to worker ranch workplaces wants biometric checks despite electronic pass. These workplaces will moreover be gotten by front line alert systems. Despite real security features as well as firewall and VPN development will have to be improved to guarantee data move. New firewall draws near, though not totally described, will reduce the VPN traffic to unequivocal IP areas and ports. With refreshed firmware, breaks are going to be less disposed to occur [11].

CONCLUSION

This research work introduced the advancement of computational cloud models, definition, qualities, design, and empowering innovations. It additionally comprises fundamental terms identified

with distributed computing with the intent to address questions often-times asked by individuals those in the PC field. In this exploration work, we also additionally depicted the benefits, drawbacks of all the various models concerning the two suppliers and customers, difficulties to embrace. At last, we have more centered around distributed computing and its nuts and bolts. In highlighting research, we will give a lot of consideration on future parts of distributed computing like how we do with secure, quick, continuous access and all this add-on makes it more profitable.

REFERENCES

1. In: Buyya R, Broberg J, Goscinski A. Cloud Computing: Principles and Paradigms: Introduction to Cloud Computing. New York, USA: Wiley; 2011.
2. Ryan Patrick S, Merchant Ronak, Falvet Sarah. Regulation of the cloud in India. J Internet Law. 2011; 15(4): 1–17.
3. Priyanshu Srivastava, Rizwan Khan. A Review Paper on Cloud Computing. Int J Adv Res Comput Sci Softw Eng. 2018; 8(6): 17–20.
4. Markus Böhm, Stefanie Leimeister, *et al.* Cloud Computing and Computing Evolution. Technische Universität München (TUM). 2011.
5. Rabi Prasad Padhy, Manas Ranjan Patra. Evolution of Cloud Computing and Enabling Technologies. International Journal of Cloud Computing and Services Science (IJ-CLOSER). 2012; 1(4): 182–198.
6. Yoo Christopher S. Cloud Computing: Architectural and Policy Implications. Rev Ind Organ. Forthcoming. 2011; 11–15.
7. Mell Peter and Grance Timothy. The NIST definition of cloud computing: recommendations of the National Institute of Standards and Technology. Special Publication. 2011; 800–145.
8. Rajleen Kaur, Amanpreet Kaur. A Review Paper on Evolution of Cloud Computing, its Approaches and Comparison with Grid Computing. International Journal of Computer Science and Information Technologies (IJCSIT). 2014; 5(5): 6060–6063.
9. Ajit Kumar. Cloud Computing in Indian Perspective. Academic Library Services Through Cloud Computing-Moving Libraries to the Web. 2013 Feb 12–15; Guru Gobind Singh Indraprastha University, New Delhi. 2013.
10. The American Bazaar (2021, Apr) United States offers to make vaccine ingredients available to India immediately. [Online]. Available from <https://www.americanbazaaronline.com/>. [Accessed Apr 25, 2021].
11. The New Indian Express (2021, Apr) SBI offers video KYC-based savings account opening feature on Yono [Online]. Available from: <https://www.newindianexpress.com/business/>. [Accessed Apr.26, 2021].